BMS COLLEGE OF ENGINEERING

(Autonomous college, affiliated to Visveswaraya Technological University)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION

V Semester

ADVANCED DIGITAL LOGIC DESIGN

(23EC5PE1AD)

AAT Report 2

Topic: Digital Logic Modules Simulations

Sl No	Student Name	USN	Student signature	Signature of the Staff In charge	Marks Obtained
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Course Instructor:

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Dept. of ECE, BMSCE

Table of Contents

1. 4-Bit Parity Generator and Checker
2. Ring Counter

Tools Used

The design, implementation, and verification of the twelve digital logic modules described in this report were conducted using an industry-standard Electronic Design Automation (EDA) software suite running within an enterprise Linux environment. The complete simulation and debugging flow relies on Cadence Design Systems software tools, providing high-performance compilation, simulation, and waveform analysis capabilities.

Linux Operating System Environment

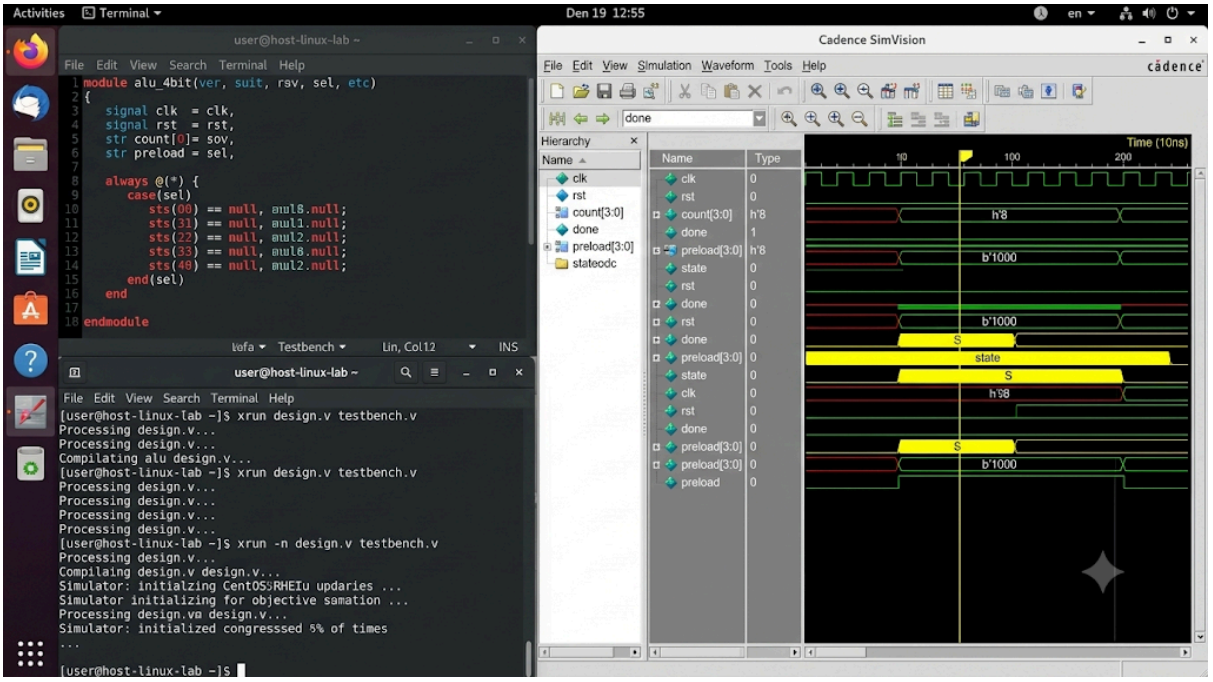
All EDA tools were deployed on a 64-bit Linux distribution, which serves as the standard operating system environment for semiconductor design and verification in enterprise and academic settings. The Linux environment provides stable process management, efficient file system handling, and robust multi-user resource allocation required for high-throughput digital logic simulations and command-line automation via shell scripting.

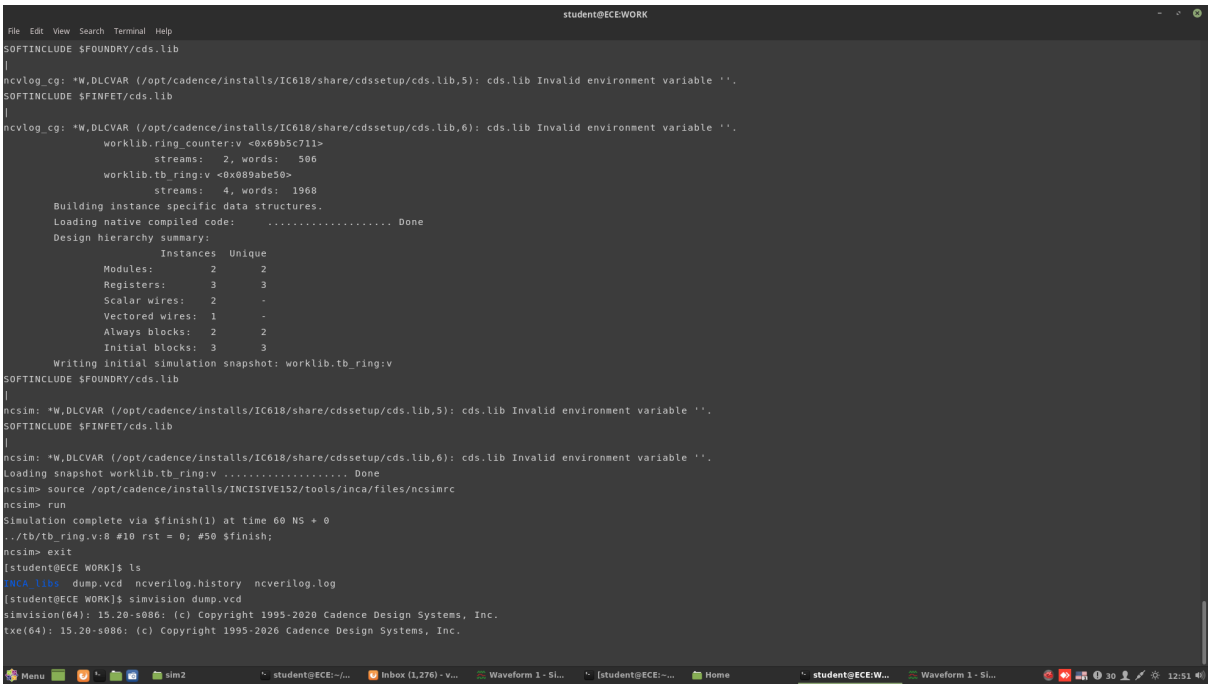
Cadence Xcelium / NC-Verilog Simulator

Logic compilation and dynamic functional simulation were performed using the Cadence logic simulation engine (NC-Verilog / Xcelium Enterprise Simulator). The compiler processes IEEE 1364-compliant Verilog HDL module descriptions alongside their corresponding testbenches. The simulator executes event-driven functional verification, enforcing precise timing control (`timescale 1ns/1ps), evaluation of logic gate delays, and signal transitions across all internal design nodes.

Cadence SimVision Waveform Viewer

Waveform analysis and visual signal verification were conducted using Cadence SimVision. During simulation execution, value change dump files (.vcd) or database files were generated and imported into SimVision to inspect time-domain logic behavior. The graphical interface enables precise timing measurements, bus signal expansion, state machine transition tracking, and glitch analysis across test vectors to validate full design correctness prior to hardware synthesis.


Activities
Terminal
Den 19 12:55
en
user@host-linux-lab ~
File Edit View Search Terminal Help
module alu_4bit(ver, suit, rsv, sel, etc)
{
signal clk = clk,
signal rst = rst,
str count[0]= sov,
str preload = sel,
always @(*) {
case(sel)
sts(00) == null, mul8.null;
sts(31) == null, mul1.null;
sts(22) == null, mul2.null;
sts(33) == null, mul8.null;
sts(40) == null, mul2.null;
end(sel)
end
endmodule
Testbench
Lin, Col 12
INS
user@host-linux-lab ~
File Edit View Search Terminal Help
[user@host-linux-lab ~]$ xrun design.v testbench.v
Processing design.v...
Processing design.v...
Compilating alu design.v...
[user@host-linux-lab ~]$ xrun design.v testbench.v
Processing design.v...
Processing design.v...
Processing design.v...
Processing design.v...
[user@host-linux-lab ~]$ xrun -n design.v testbench.v
Processing design.v...
Compilaing design.v design.v...
Simulator: initialzing CentOS5RHEIu updaries ...
Simulator initializing for objective samation ...
Processing design.va design.v...
Simulator: initialized congresssed 5% of times
...
[user@host-linux-lab ~]$
Cadence SimVision
File Edit View Simulation Waveform Tools Help
cadence
done
Hierarchy
Name
clk
rst
count[3:0]
done
preload[3:0]
stateodc
Name
Type
clk 0
rst 0
count[3:0] h'8
done 1
preload[3:0] h'8
state 0
rst 0
done 0
rst 0
done 0
preload[3:0] 0
state 0
clk 0
rst 0
done 0
preload[3:0] 0
preload[3:0] 0
preload 0
Time (10ns)
100
200
h'8
b'1000
b'1000
S
state
S
h'8
S
b'1000


student@ECE:WORK
File Edit View Search Terminal Help
SOFTINCLUDE $FOUNDRY/cds.lib
|
ncvlog_cg: *W,DLCVAR (/opt/cadence/installs/IC618/share/cdssetup/cds.lib,5): cds.lib Invalid environment variable ''.
SOFTINCLUDE $FINFET/cds.lib
|
ncvlog_cg: *W,DLCVAR (/opt/cadence/installs/IC618/share/cdssetup/cds.lib,6): cds.lib Invalid environment variable ''.
worklib.ring_counter:v <0x69b5c711>
streams: 2, words: 506
worklib.tb_ring:v <0x089abe50>
streams: 4, words: 1968
Building instance specific data structures.
Loading native compiled code: ..... Done
Design hierarchy summary:
Instances Unique
Modules: 2 2
Registers: 3 3
Scalar wires: 2 -
Vectored wires: 1 -
Always blocks: 2 2
Initial blocks: 3 3
Writing initial simulation snapshot: worklib.tb_ring:v
SOFTINCLUDE $FOUNDRY/cds.lib
|
ncsim: *W,DLCVAR (/opt/cadence/installs/IC618/share/cdssetup/cds.lib,5): cds.lib Invalid environment variable ''.
SOFTINCLUDE $FINFET/cds.lib
|
ncsim: *W,DLCVAR (/opt/cadence/installs/IC618/share/cdssetup/cds.lib,6): cds.lib Invalid environment variable ''.
Loading snapshot worklib.tb_ring:v ..... Done
ncsim> source /opt/cadence/installs/INCISIVE152/tools/inca/files/ncsimrc
ncsim> run
Simulation complete via $finish(1) at time 60 NS + 0
../tb/tb_ring.v:8 #10 rst = 0; #50 $finish;
ncsim> exit
[student@ECE WORK]$ ls
INCA_libs dump.vcd ncverilog.history ncverilog.log
[student@ECE WORK]$ simvision dump.vcd
simvision(64): 15.20-s086: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
txe(64): 15.20-s086: (c) Copyright 1995-2026 Cadence Design Systems, Inc.

1. 4-Bit Parity Generator and Checker (Combinational Circuit)

Design

- **Generator:** Uses XOR cascades to create an even parity bit. Generates `parity_bit` based on `d`.
- **Checker:** Evaluates a 5-bit input (4-bit data + 1-bit parity) to output `error_detected` if parity is odd.

Circuit Diagram & Truth Table

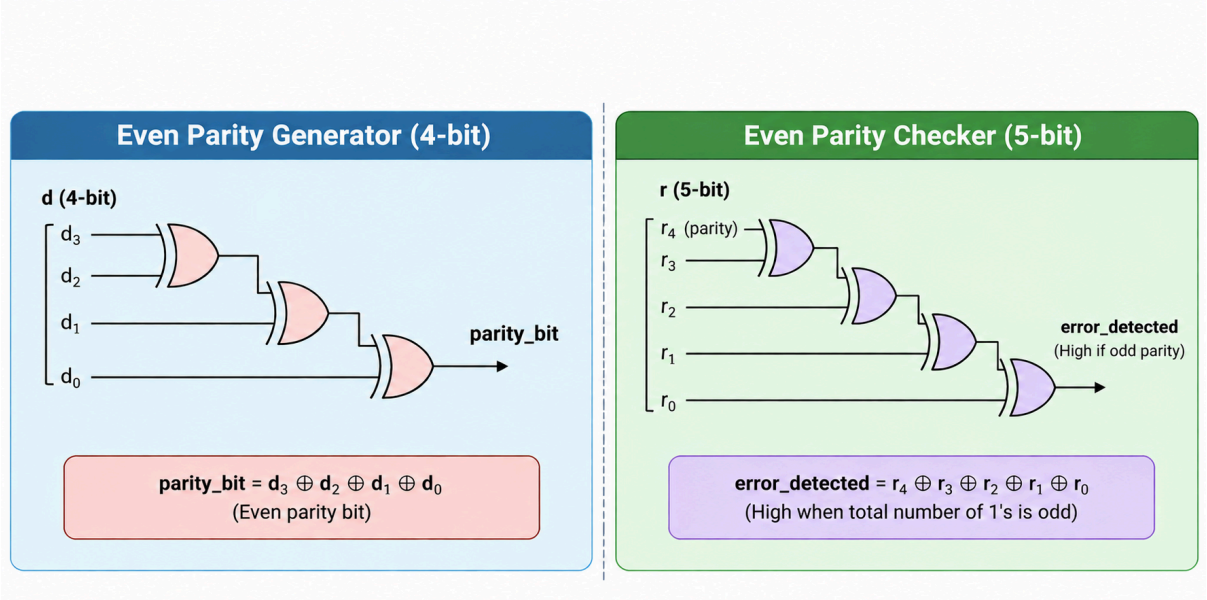


Figure 3.1: Even Parity Generator and Checker Block Diagrams

Truth Tables for Even Parity Generator and Checker (4-bit)

Even Parity Generator (4-bit)

Data Input (d)				Output
d_3	d_2	d_1	d_0	parity_bit
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Even Parity Checker (5-bit)

Received Input (r)					Output
r_4 (parity)	r_3	r_2	r_1	r_0	error_detected
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	1
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	0	1	0
0	0	1	1	0	0
0	0	1	1	1	1
0	1	0	0	0	1
0	1	0	0	1	0
0	1	0	1	0	0
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	0	1	1
0	1	1	1	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	0	0	1	0
1	0	0	1	0	0
1	0	0	1	1	1
1	0	1	0	0	0
1	0	1	0	1	1
1	0	1	1	0	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	0	1	1
1	1	0	1	0	1
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	0	1	0
1	1	1	1	0	1
1	1	1	1	1	0

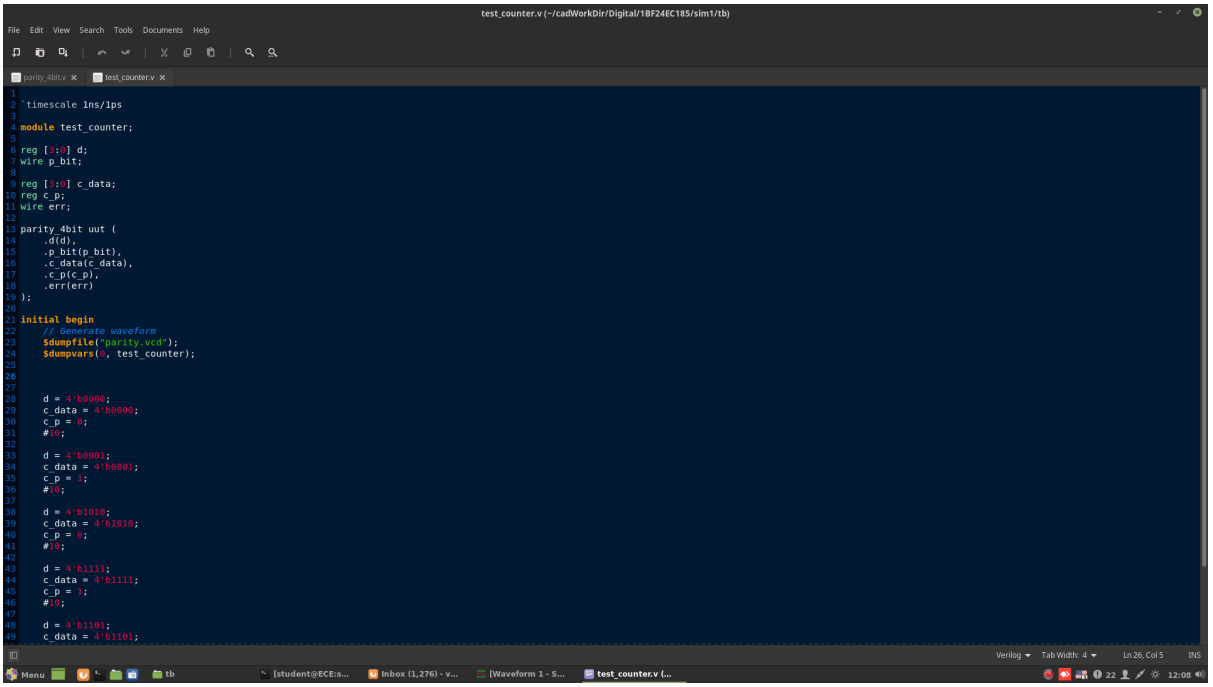
Table 3.1: Truth Tables for Even Parity Generator and Checker

Verilog HDL Code

File Edit View Search Tools Documents Help

```
1  `timescale 1ns/1ps
2
3  module parity_4bit (
4      input [3:0] d,
5      output p_bit,
6      input [3:0] c_data,
7      input c_p,
8      output err
9  );
10
11 assign p_bit = d[0] ^ d[1] ^ d[2] ^ d[3];
12
13 assign err = c_data[0] ^ c_data[1] ^
14             c_data[2] ^ c_data[3] ^ c_p;
15
16 endmodule
```

Testbench



Output

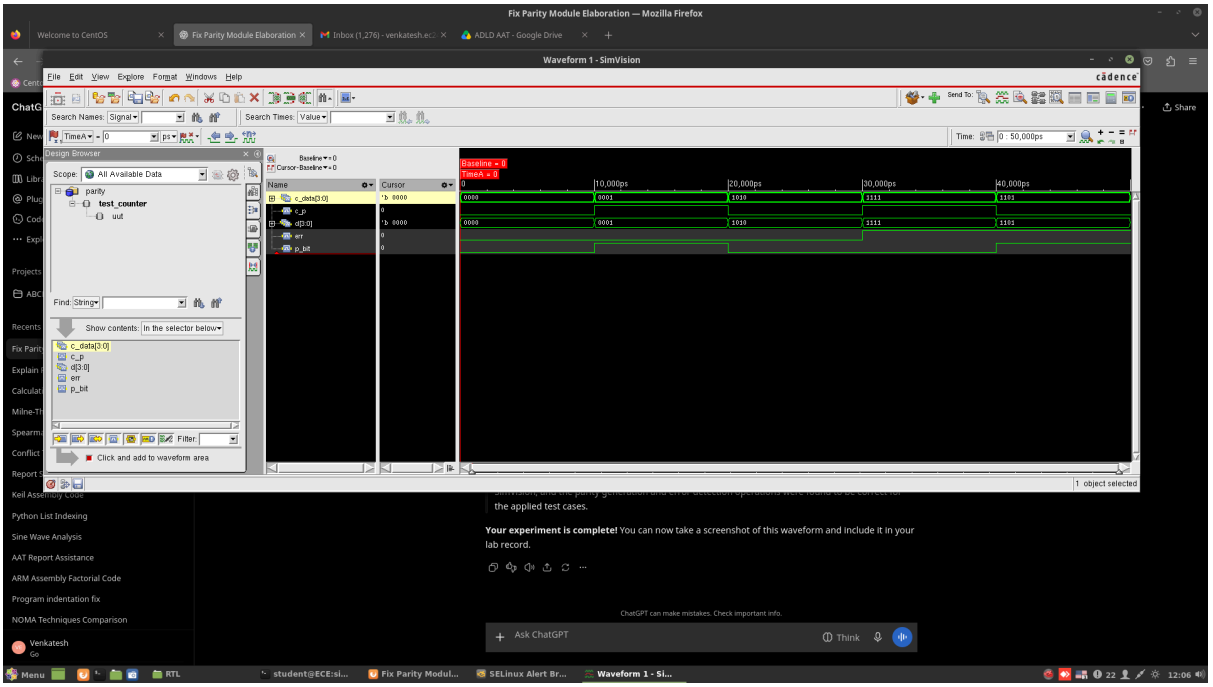


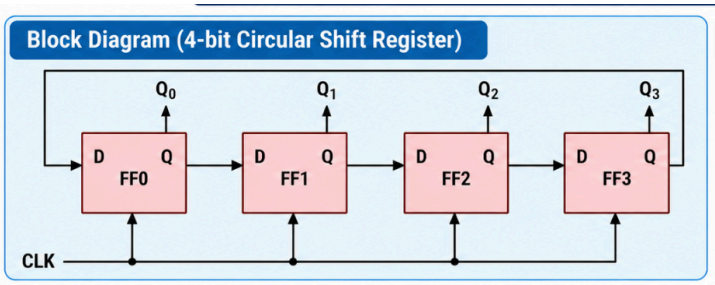
Figure 3.2: Parity Generator and Checker Simulation Waveform

2. Ring Counter(Sequential Circuit)

Design

A sequential shift register where the last flip-flop feeds back into the first. This recirculates a single `'1'` (or `'0'`) pattern in a loop.

Circuit Diagram & Truth Table



State Table / Sequence (Single '1' Circulating)

Clock Cycle	Outputs (Q_3 Q_2 Q_1 Q_0)			
Initial	0	0	0	1
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
...	(repeats)			

Figure 6.1: Ring Counter Block Diagram and State Sequence Table

Verilog HDL Code

File Edit View Search Tools Documents Help

```
1 module ring_counter (input clk, rst, output reg [3:0] q);
2 always @(posedge clk or posedge rst)
3 if (rst) q <= 4'b0001;
4 else q <= {q[0], q[3:1]};
5 endmodule
```

Testbench

File Edit View Search Tools Documents Help

ring_counter.v tb_ring.v

```
1 module tb_ring;
2 reg clk=0, rst=1; wire [3:0] q;
3 ring_counter uut (clk, rst, q);
4 always #5 clk = ~clk;
5 initial begin
6 $dumpfile("dump.vcd");
7 $dumpvars(0, tb_ring);
8 #10 rst = 0; #50 $finish;
9 end
10 endmodule
```

Output



Figure 6.2: Ring Counter Simulation Waveform